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A case report of bacteriophage therapy for the treatment of lung infection due to carbapenem-resistant *Pseudomonas aeruginosa*

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In recent years, with the increasing global antimicrobial resistance, bacteriophage therapy has emerged as one of the most promising alternatives to antibiotics. Numerous studies have reported the effectiveness and safety of phage therapy in treating multidrug-resistant bacterial infections. However, there is limited research on the relationship between the emergence of phage-resistant strains and clinical outcomes. In this study, we applied phage nebulization therapy to a patient with carbapenem-resistant *Pseudomonas aeruginosa* lung infection. After 8 weeks of treatment with *P. aeruginosa* phages, the patient's sputum bacterial load decreased, overall inflammatory markers decreased, and lung effusion significantly reduced despite isolation of phage-resistant strains. Whole-genome sequencing of resistant isolates, coupled with *Galleria mellonella* larvae infection modeling, revealed a trade-off effect, demonstrating attenuated virulence associated with phage resistance. This finding underscores the potential for phage therapy to achieve positive clinical outcomes even in the face of resistance development, highlighting the need for a nuanced understanding of phage-bacteria coevolution in clinical settings.

Keywords *Pseudomonas aeruginosa*, Multidrug-resistant bacteria, Lung infection, Bacteriophage therapy

Pulmonary infections caused by *Pseudomonas aeruginosa* pose a significant therapeutic challenge, particularly in the context of increasing antibiotic resistance and the emergence of multidrug-resistant strains[1,2]. Treating *P. aeruginosa* infections is extremely difficult due to its rapid mutation and adaptation to acquire antibiotic resistance. Additionally, the ability of *P. aeruginosa* to form biofilms hinders infection treatment, as biofilms serve as diffusion barriers that restrict the entry of antibiotics into bacterial cells[3]. The World Health Organization (WHO) has published a list of bacteria that urgently need new antibiotics. Notably, carbapenem-resistant *Pseudomonas aeruginosa* (CRPA) has been designated as a high priority pathogen in the list of antibiotic-resistant bacteria[4]. There is an urgent need to explore new treatment methods for infections caused by multidrug-resistant *P. aeruginosa*[5]. Bacteriophages are viruses that can infect and replicate within bacterial cells, and they can target and kill bacteria rapidly and selectively. They are considered one of the most promising alternatives in the face of the antibiotic crisis[6–9].

In the field of phage therapy, lytic phages are the primary modality for clinical applications due to their inherent ability to directly lyse host bacteria. These phages efficiently infect bacterial cells, undergo extensive replication, and ultimately lyse the host to release progeny phages, demonstrating unequivocal bacteriolytic effects. According to the clinical guidelines of the International Phage Therapy Association (IPTA), lytic phages

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can be further classified into three major families: Myoviridae (characterized by contractile tails), Siphoviridae (possessing long non-contractile tails), and Podoviridae (featuring short non-contractile tails). Phages belonging to the family myovirus possess contractile tails with retractable sheaths, exhibiting not only potent lytic activity but also the remarkable ability to penetrate bacterial biofilms - features that confer significant therapeutic advantages[10]. In contrast, siphovirus phages are characterized by their long, non-contractile tail structures, demonstrating high adsorption efficiency to host bacteria, broad host range, and genomic conservatism that collectively reduce the risk of lysogeny and enhance treatment safety[11]. Meanwhile, podovirus phages feature short, non-contractile tails while maintaining strong lytic capabilities; however, they exhibit a relatively narrow host range and pronounced host specificity. In contrast, temperate phages are strictly excluded from clinical use due to their potential to integrate into the host bacterial genome as prophages, which may facilitate horizontal gene transfer of antibiotic resistance genes and virulence factors. This risk underscores the critical necessity of prophage detection in therapeutic phage preparations prior to clinical application[12–14].

Phage therapy has a history of one hundred year in treating infections caused by antibiotic-resistant bacteria and has shown significant efficacy in successfully treating life-threatening multidrug-resistant bacterial infections[15–18]. However, there is relatively limited research on the factors that affect the success or failure of phage therapy, particularly the emergence of phage resistance. Here, we report the case of a severely ill pneumonia patient who showed significant improvement after receiving nebulized lytic phage therapy, and we explore the relationship between the emergence of phage-resistant strains and the clinical outcome.

Results

Patient clinical history

On August 14, 2022, an 83-year-old male patient was admitted to the hospital due to "sudden altered consciousness for more than 2 hours". The patient denied any history of "coronary heart disease, diabetes, kidney disease", or other underlying medical conditions. Emergency head CT did not reveal any cerebral hemorrhage or extensive cerebral infarction. Following admission, the patient received treatment including oxygen therapy, ceftriaxone for infection control, and red blood cell transfusion for anemia improvement. The patient's respiratory condition gradually deteriorated, presenting symptoms of respiratory distress and decreasing oxygen saturation (SPO2), ultimately leading to a diagnosis of interstitial pneumonia and acute respiratory distress syndrome. In the early morning of August 17, 2022, the patient's respiratory distress worsened, and SPO2 could not be maintained. After bronchoscopy treatment with endotracheal intubation, the patient was transferred to a ventilator for respiratory support. During the patient's hospitalization, he experienced recurrent lung infections with multiple pathogens, including multidrug-resistant (MDR) *Klebsiella pneumoniae*, MDR *P. aeruginosa*, and *Candida albicans*. Antibiotic treatments were administered based on sensitivity testing results, including cefoperazone/sulbactam, meropenem, ciprofloxacin, amikacin, isepamicin, minocycline, ceftazidime-avibactam, colistin, piperacillin/tazobactam, and fluconazole (see more details in Supplementary Table S1), but were unsuccessful in eradicating the pulmonary pathogens. On February 16, 2023, sputum culture revealed CRPA, (Fig. 1A) which was only sensitive to colistin and tobramycin (Supplementary Table S2) according to sensitivity testing. At the same time, the patient's C-reactive protein (CRP) levels reached 150 mg/L (Fig. 1E), indicating severe illness. Due to the critical condition of the patient, with the approval of the Ethics Committee of Southern University of Science and Technology Hospital and the consent of the patient's family, phage therapy was initiated.

Phage therapy and clinical course following

The clinical *P. aeruginosa* isolate (NPA09B) obtained from the patient's sputum was sent to the laboratory at the Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences for phage screening. The phage activity was validated through an in vitro lysis curve, and phages PaSz-1_45_92k and PaZh_1 isolated from different sewage samples showed high lytic activity against the patient's *P. aeruginosa* isolate. Under conditions of multiplicity of infection (MOI)0.1, our phage cocktail was able to effectively inhibit bacterial growth for 12 h (Fig. 2G and J). Starting from February 26, 2023, personalized phage therapy was initiated. During the initial phage therapy course, the patient's inflammatory markers remained elevated, possibly due to the infection entering a severe stage, though there was no progressive deterioration in the patient's condition. Using the multidrug-resistant *P. aeruginosa* strain (NPA10B) as the host, we isolated and cultured a phage PAL9 capable of lysing strain NPA10B, which led us to adjust the formulation of our phage cocktail. Phage cocktail_2 is composed of PaSz-1_45_92k, PaZh_1, and PAL9, with each phage at a titer of 3×10^9 PFU/ml, resulting in a titer of 9×10^9 PFU/ml for the phage cocktail. (Table 1). To ensure effective clearance of the resistant bacteria, subsequent treatment continued with the same phage cocktail. On days 4, 20, 32, and 43 post-phage therapy initiation, *P. aeruginosa* was still cultured from the patient's sputum, albeit with decreased bacterial load (Supplementary Table S3). The patient's fluid intake, heart rate, and blood pressure remained stable, while blood gas parameters such as partial pressure of oxygen in arterial blood (PaO_2), partial pressure of arterial blood carbon dioxide ($PaCO_2$), and lactate improved significantly compared to earlier measurements. Before treatment(D-7), the patient required continuous infusion of norepinephrine at a dose of 0.15 μg/kg/min to maintain blood pressure at 96/51 mmHg. At an inspired oxygen fraction (FiO_2) of 0.6, the PaO_2 was only 69 mmHg, while the $PaCO_2$ was as high as 57 mmHg, indicating severe hypoxemia and ventilatory dysfunction. After two weeks of phage therapy (D14), the patient showed significant clinical improvement: after discontinuation of vasopressors, blood pressure remained stably maintained at 132/67 mmHg. When FiO_2 was reduced to 0.5, PaO_2 increased to 123 mmHg, while $PaCO_2$ decreased from 57 mmHg to 44 mmHg, demonstrating marked recovery of both pulmonary oxygenation and ventilation function.

Infection markers like white blood cells, neutrophils, procalcitonin and hypersensitive CRP gradually normalized (Fig. 1B and E), suggesting *P. aeruginosa* colonization rather than active infection. The patient continued intermittent off-ventilator exercises, maintaining a stable condition thereafter. On the 90th-day post-

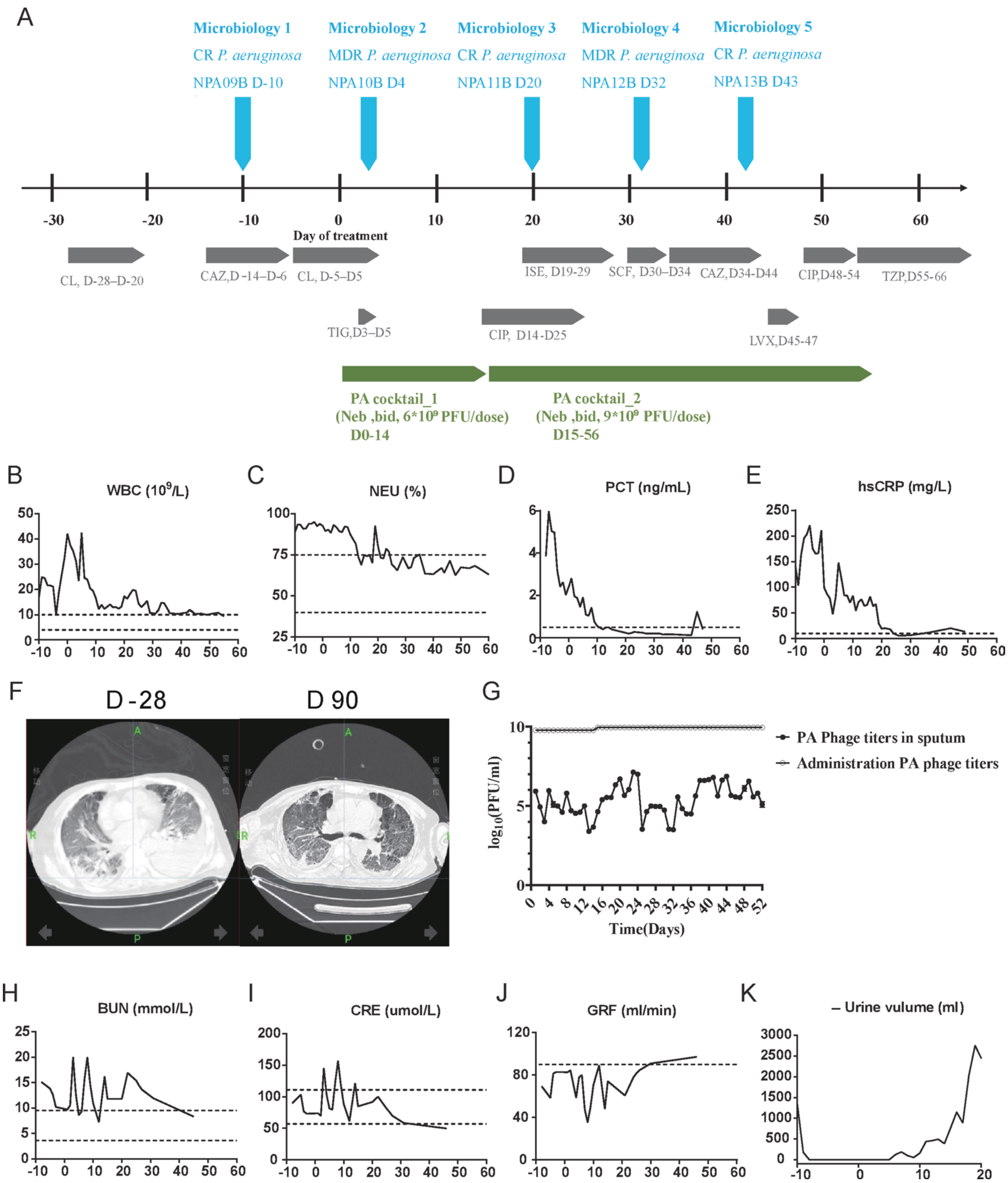

Fig. 1. Clinical data. (**A**) Timeline of microbiological results (light blue), antibiotic therapies (dark gray), and phage therapy (blue-green). Samples are numbered concerning the first phage administration (D0). Sputum samples were obtained 2–3 h after phage administration. TGC, Tigecycline; TZP, Piperacillin/Tazobactam; CAZ, Ceftazidime; SCF, Cefoperazone/Sulbactam; FEP, Cefepime; CIP, Ciprofloxacin; LEV, Levofloxacin; CL, Colistin. ISE, Isepamicin. (**B–E**) Patient's inflammatory factor profile. No major alterations were observed (day 0 indicates the time when phage therapy was initiated). WBC, mean white blood cells; NEU, neutrophils; PCT, procalcitonin; Hs-CRP, high-sensitivity C-reactive protein. Normal ranges are presented between dotted lines in WBC and NEU; normal ranges are below the dotted line for PCT and hs-CRP. (**F**) The patient's chest CT scans 28 days before phage therapy and on day 90 after phage therapy. (**G**) Administrated phage titers and phage titers in sputum detected by plaque assays following the phage therapy. The left Y-axis represents phage titers. (**H–K**) Renal function indicators change. Normal ranges are presented between dotted lines in BUN and CRE; normal ranges are below the dotted line for GRF. BUN: Blood Urea Nitrogen; CRE: Creatinine; GFR: Glomerular Filtration Rate.

phage therapy initiation, a computed tomography scan revealed a significant reduction in pleural effusion in both lungs, particularly on the left side, with partial lung tissue expansion and incomplete re-expansion (Fig. 1F). Analysis of sputum samples collected during treatment showed phage activity levels reaching up to 10^7 PFU/ml in the lungs post-nebulization (Fig. 1G). Renal function indices such as blood urea nitrogen, blood creatinine, glomerular filtration rate and urine output of patient in the early stage of phage therapy were at abnormal levels

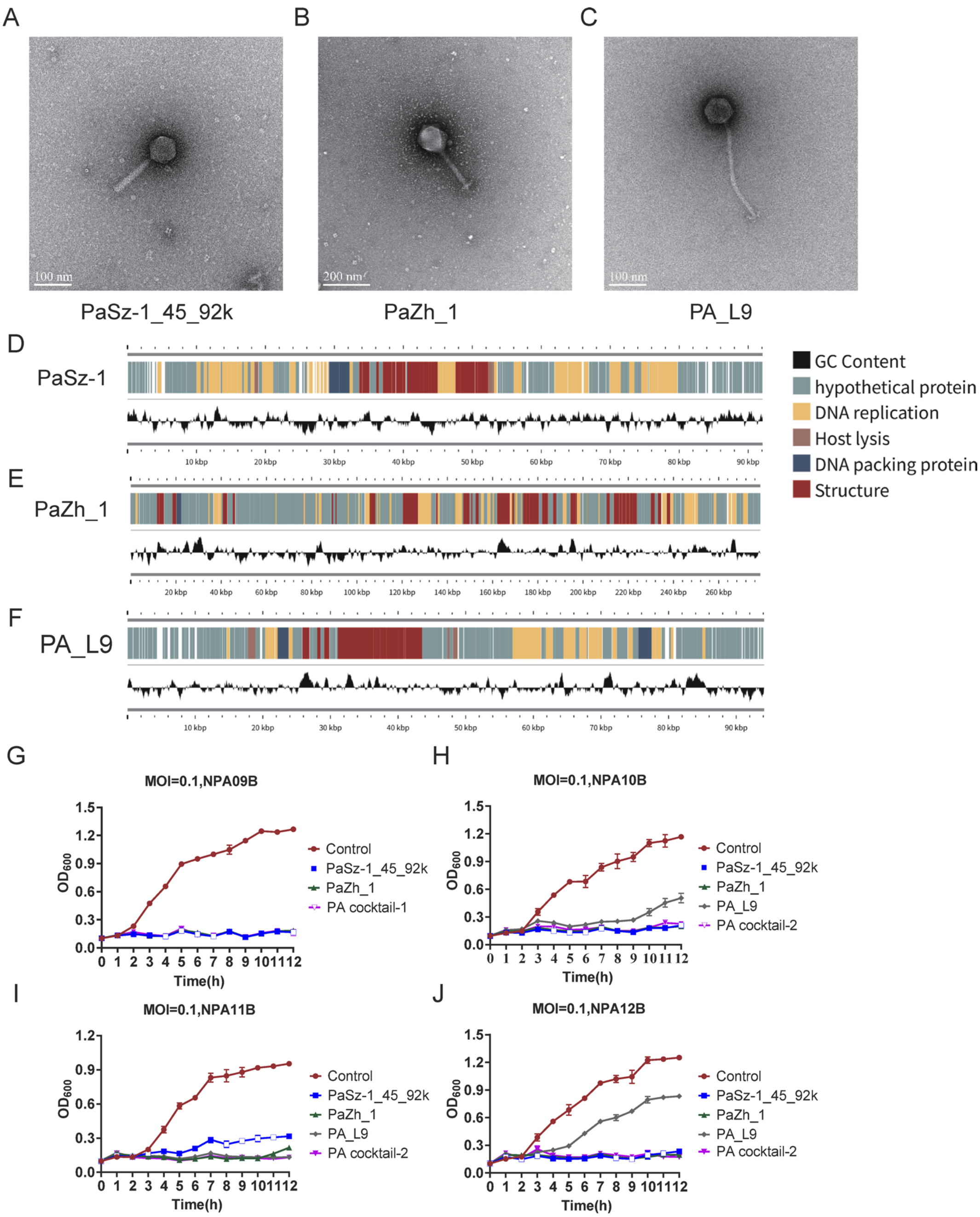

Fig. 2. Phage characterization and phage inhibition of bacteria under liquid culture conditions. (**A–C**) Transmission electron micrograph of the phages. Scale bar represents 100 nm and 200 nm, PaSz-1_45_92k, PaZh_1 and PAL9. (**D–F**) Phage genome mapping. The colors of ORFs refer to five modules: phage structure, red; host lysis, brown; DNA packaging, blue; DNA replication, yellow; hypothetical proteins, Morandi green; and GC content, black. (**G–J**) Activities of single phage (at the multiplicity of infection [MOI] = 0.1), as well as phage cocktail against NPA09B, NPA10B, NPA11B and NPA12B were determined using a Microplate Reader. Results are presented as mean values of three experiments (biological replicates) with error bars representing the standard deviations of the means.

(Fig. 1H and K), and patient was being treated with colistin for anti-infective therapy at this stage, which might be related to the nephrotoxicity of colistin. After discontinuing colistin, the patient's renal function indices gradually returned to normal levels, and the patient demonstrated good tolerance with no significant phage-related adverse reactions, such as allergy.

Phage characterization

Transmission electron microscopy revealed that PaSz-1_45_92k and PaZh_1 belong to the myovirus (Fig. 2A and B), PAL9 belongs to the family of siphovirus (Fig. 2C). The heads of all three phages were icosahedral, and

Phage cocktail	Phage	Titer (PFU/dose)	Administration time (Days)	Administration method and frequency
PA cocktail_1	PaSz-1_45_92k	$6*10^9$	14	Nebulization Bid
	PaZh_1			
PA cocktail_2	PaSz-1_45_92k	$9*10^9$	42	
	PaZh_1			
	PA_L9			

Table 1. Composition of phage cocktail and mode of administration.

the head diameter of phage PaSz-1_45_92k was 80 nm, and the tail length was 102 nm; the head diameter of phage PaZh_1 was 120 nm, and the tail length was 200 nm; the head diameter of phage PAL9 was 80 nm and 300 nm in tail length (Supplementary Table S4). The sequences of the three phages composing the phage cocktail have been determined and deposited in the GenBank database: PaSz-1_45_92k (GenBank: MN871481), PaZh_1 (GenBank: MN871489) and PA_L9 (GenBank: PP869205). The lengths of the three phage genomes are PaSz-1_45_92k: 92,096 bp; PaZh_1: 278,407 bp and PAL9: 94,141 bp. Through sequence analysis, PaSz-1_45_92k, PaZh_1, and PAL9 are classified into the genera *Pakpunavirus*, *Phikzvirus*, and *Samunavirus*, respectively. The genomic maps of the three phages are shown in Fig. 2D and F. All three phages possess a range of genes encoding common phage-related features. Genome analysis of the three phages did not reveal any toxins, antibiotic resistance, or genes encoding lysogenic determinants.

In addition, we also determined the in vitro bacteriolytic effects of these three phages and phage cocktails on isolates before phage treatment (NPA09B) and isolates during phage treatment (NPA10B, NPA11B, NPA12B, and NPA13B). The phage cocktail had the best bacterial inhibition at the same MOI, and phage-resistant strains were more likely to emerge using a single phage (Fig. 2G and J). One of these five isolates (NPA13B) was resistant to all three phages in the phage cocktail. We also tested this strain for susceptibility to other *P. aeruginosa* phages and did not identify phages that could lyse this strain (Supplementary Fig. 1).

Genome analysis of clinical *P. aeruginosa* isolates

Phages not only lyse bacteria but can also indirectly threaten bacterial proliferation by imposing selective pressure. The development of phage-resistant strains requires genetic variation. Five clinical *P. aeruginosa* strains, NPA09B, NPA10B, NPA11B, NPA12B and NPA13B were sequenced to closure using a combination of short-read (Illumina) and long-read (PacBio) sequencing to investigate pathogen evolution during the course of phage treatment. The bacterial chromosome lengths among the strains exhibit minor differences, and multi-locus sequence typing (MLST) reveals that all five strains belong to sequence type ST260. These isolates represent the evolution of strains from a common ancestor over time rather than a succession invasion by different strains. Consistent with the broad antibiotic resistance observed in these isolates, 59 antibiotic resistance genes (ARGs) were identified based on searches against the CARD 2023 database (Supplementary Table S5). More single nucleotide polymorphisms (SNPs) and (indels) were observed between these isolates. Summaries of the genomes and changes observed in clinical isolates (relative to NPA09B) are summarized in (Table 2), and the locations of prophages and genomic islands, in NPA09B genome, are illustrated in Fig. 3. In reference to NPA09B, detailed sequence changes, associated coordinates and genes affected in other clinical isolates are listed in Table 2, respectively. Notably, in the NPA12B isolate, the 18 kbp polysaccharide synthesis locus (Psl) gene cluster[19]which encodes biofilm-forming related proteins, was deleted (Fig. 3). In addition, genes involved in antibiotic resistance, biofilm formation, synthesis of lipopolysaccharide (LPS), development of Type IV pili, and pyrimidine metabolism were also detected. See more detail in Table 2.

The CCV algorithm selected strain NPA09B as the reference (inner circle). Moving from the inner to outer rings of the CCV, there is a light blue ring, a purple ring, and a teal ring displaying genomic variations in NPA10B, NPA11B, and NPA12B, while an outer gray-green ring shows genomic variations in the isolate NPA13B. The two multi-colored outer rings display protein annotations (categories) sourced from the Clusters of Orthologous Groups of proteins (COGs) database.

Clinical strains characterization

The efficacy of plating (EOP) of phage Pa-Sz-1 on NPA10B, NPA11B, and NPA12B decreased by ~1 $\log_{10}$ compared to NPA09B (Fig. 4A), indicating slight phage resistance. Notably, NPA13B showed no sensitivity to the tested phages, suggesting complete phage resistance (Fig. 4A and Supplementary Fig. S1). In addition, the adsorption rate of phage PaSz-1 on NPA13B was significantly reduced compared to the pre-treatment strain NPA09B, suggesting that surface structure modifications in NPA13B block phage adsorption (Fig. 4B). The development of phage-resistant strains with genetic variation, may lead to the loss of crucial survival traits in bacteria, such as virulence or antibiotic resistance[20,21]. We evaluated the antibiotic susceptibility profiles of five clinical isolates. The results demonstrated that the minimum inhibitory concentrations (MICs) of four phage-treated isolates (NPA10B, NPA11B, NPA12B, and NPA13B) exhibited significant alterations compared to their parental strain NPA09B. All clinical isolates showed a decreased MIC for β-lactams and an increased MIC for quinolones (Fig. 4C and Supplementary Table S2). The common factor among all isolates was a G154R point mutation in the *ampR* gene (Table 2), which is known to positively regulate β-lactam resistance and negatively regulate quinolone resistance[22,23]. This mutation contributes to the observed enhancement of β-lactam resistance and reduced quinolone resistance[24]. In particular, NPA10B also carries a frameshift mutation in the *nuoN* gene, which leads to the overexpression of *ampC* through *ampR*, further increasing β-lactam resistance[25].

	Mutation type	Position in NPA09B	Length (bp)	Mutation	Affected genes	Functions	Previously reported effects of similar mutations	References
NPA10B	snp	433,070	1	"A" to "G" (Lys111 to Glu)	gene encodes hypothetical protein	unkown	/	
	snp	469,714	1	"T" to "A" (Leu100 to Gln)	*rstB*	regulation expression of several efflux pumps involved in multidrug resistance	/	
	snp	537,023	1	"C" to "A" (Pro22 to Thr)	*str*	resistance to streptomycin	/	
	snp	853,703	1	"A" to "C" (Tyr33 to Ser)	gene encodes hypothetical protein	unkown	/	
	Deletion	942,392	9	"TCCTCCTGC" deletion in frame (deletion Ile51_Arg54, insertion Ser)	*mexZ*	negative regulation of the MexXY-OprM efflux pump associated with aminoglycoside and fluoroquinolone resistance	/	
	Insertion	1,622,892	2	"TC" insertion in frame (frameshift mutation from Lys394)	*nuoN*	regulation resistance to β-lactams through ampR	mutation in the nuoN gene is linked to increased β-lactam resistance, however with mutation in the ampR gene leads to decreased β-lactam resistance	19
	Insertion	1,631,821	2	"AG" insertion in frame (frameshift mutation from Ser323)	*bqsS*	regulation genes expression in response to iron levels, influencing biofilm formation and virulence	/	
	snp	3,208,074	1	"C" to "G" (Arg154 to Gly)	*ampR*	postive regulation resistance to β-lactams and negative regulation resistance to quinolones	mutation in the ampR gene is linked to decreased β-lactam resistance	17
NPA11B	snp	433,070	1	"A" to "G" (Lys111 to Glu)	gene encodes hypothetical protein	unkown		
	Deletion	942,436	1	"T" deletion in frame (frameshift mutation from Val39)	*mexZ*	negative regulation of the MexXY-OprM efflux pump associated with aminoglycoside and fluoroquinolone resistance	/	
	snp	2,188,265	1	"C" to "T" (Thr83 to Ile)	*gyrA*	cleaves and rebinds double-stranded host DNA	mutation in the gyrA gene is linked to resistance to quinolone-based antibiotics	20
	Deletion	2,287,240	11	"GCTCGACGGTG" deletion in frame (frameshift mutation from Arg112)	*actA*	iron uptake	/	
	snp	2,433,927	1	"T" to "A" (Trp598 to Arg)	*nosR*	regulation expression of the nitrous-oxide reductase	/	
	Deletion	2,779,039	8	"ACCGAACG" deletion in frame (frameshift mutation fromThr9)	*nalC*	regulation expression of several efflux pumps involved in multidrug resistance	mutations in NalC induce MexAB-OprM overexpression resulting in high level of aztreonam resistance	21
	snp	3,208,074	1	"C" to "G" (Arg154 to Gly)	*ampR*	postive regulation resistance to β-lactams and negative regulation resistance to quinolones	mutation in the ampR gene is linked to decreased β-lactam resistance	17
	snp	3,686,204	1	"C" to "A" (Pro106 to Gln)	*mexR*	regulation expression of several efflux pumps involved in multidrug resistance	/	
	Deletion	3,865,669	9	"GAGAGCCTG" deletion in frame (deletion Glu925_Leu927)	*vgrG2b*	a trans-kingdom toxin that can target both bacterial and eukaryotic cells	/	
	Insertion	4,890,595	1	"G" insertion in frame (frameshift mutation from Ala28)	*wapH*	LPS capped with O antigen biosynthesis	less virulent in vivo	25
	snp	4,894,057	1	"C" to "T" (Gln8 to stop codon, nosense mutation)	*ssg*	LPS capped with O antigen biosynthesis	less motility	26
	snp	5,605,723	1	"C" to "T" (Thr139 to Met)	*ampD*	beta-lactamase expression regulator	/	
	Deletion	6,069,210	18	"CGCCGCGCGCCGACCTG C" deletion in frame (deletion Pro193_Leu198)	*ampDh3*	beta-lactamase expression regulator	/	
	Deletion	6,228,926	1	"A" deletion in frame (frameshift mutation from Thr161)	*oprD*	outer-membrane porin that can facilitate the diffusion of specific substrates, such as basic amino acids and carbapenem antibiotics	oprD deficiency confers resistance to carbapenems, especially imipenem.	22

Continued

	Mutation type	Position in NPA09B	Length (bp)	Mutation	Affected genes	Functions	Previously reported effects of similar mutations	References
NPA12B	snp	433,070	1	"A" to "G" (Lys111 to Glu)	gene encodes hypothetical protein	unkown	/	
	snp	469,714	1	"T" to "A" (Leu100 to Gln)	*rstB*	regulation expression of several efflux pumps involved in multidrug resistance	/	
	snp	537,023	1	"C" to "A" (Pro22 to Thr)	*str*	resistance to streptomycin	/	
	deletion	613,289	18,151	deletion polysaccharide synthesis locus gene cluster	polysaccharide synthesis locus gene cluster	biofilm formation	/	
	snp	853,703	1	"A" to "C" (Tyr33 to Ser)	gene encodes hypothetical protein	beta-lactamase expression regulator	/	
	Deletion	942,392	9	"TCCTCCTGC" deletion in frame (deletion Ile51_Arg54, insertion Ser)	*mexZ*	negative regulation of the MexXY-OprM efflux pump associated with aminoglycoside and fluoroquinolone resistance	/	
	Deletion	1,870,813	11	"GCCTGCCGGCG" deletion in frame (frameshift mutation from Gly128)	gene encodes a probable two-component response regulator	unkown	/	
	snp	3,208,074	1	"C" to "G" (Arg154 to Gly)	*ampR*	postive regulation resistance to β-lactams and negative regulation resistance to quinolones	mutation in the ampR gene is linked to decreased β-lactam resistance	17
	Deletion	5,966,049	1	"A" deletion in frame (frameshift mutation from Asn162)	*migA*	LPS with uncapped core oligosaccharide biosynthesis	mutation causes more capped core oligosaccharide and related with resistance to colistin	23
Continued								

	Mutation type	Position in NPA09B	Length (bp)	Mutation	Affected genes	Functions	Previously reported effects of similar mutations	References
NPA13B	snp	433,070	1	"A" to "G" (Lys111 to Glu)	gene encodes hypothetical protein	unkown	/	
	Deletion	942,436	1	"T" deletion in frame (frameshift mutation from Val39)	*mexZ*	negative regulation of the MexXY-OprM efflux pump associated with aminoglycoside and fluoroquinolone resistance	/	
	snp	2,188,265	1	"C" to "T" (Thr83 to Ile)	*gyrA*	cleaves and rebinds double-stranded host DNA	mutation in the gyrA gene is linked to resistance to quinolone-based antibiotics	20
	Deletion	2,287,240	11	"GCTCGACGGTG" deletion in frame (frameshift mutation from Arg112)	*actA*	iron uptake	/	
	snp	2,433,927	1	"T" to "A" (Trp598 to Arg)	*nosR*	regulation expression of the nitrous-oxide reductase	/	
	Deletion	2,779,039	8	"ACCGAACG" deletion in frame (frameshift mutation fromThr9)	*nalC*	regulation expression of several efflux pumps involved in multidrug resistance	mutations in NalC induce MexAB-OprM overexpression resulting in high level of aztreonam resistance	21
	snp	3,208,074	1	"C" to "G" (Arg154 to Gly)	*ampR*	postive regulation resistance to β-lactams and negative regulation resistance to quinolones	mutation in the ampR gene is linked to decreased β-lactam resistance	17
	Deletion	3,661,934	3	"GAA" deletion in frame (deletion Glu185_Asn186, insertion Asp)	*dht*	pyrimidine metabolism	/	
	snp	3,686,204	1	"C" to "A" (Pro106 to Gln)	*mexR*	regulation expression of several efflux pumps involved in multidrug resistance	/	
	Deletion	3,865,669	9	"GAGAGCCTG" deletion in frame (deletion Glu925_Leu927)	*vgrG2b*	a trans-kingdom toxin that can target both bacterial and eukaryotic cells	/	
	Insertion	4,836,150	5	"TGCCC" insertion in frame (frameshift mutation from Gln149)	*pilQ*	pilus synthesis	less motility, less virulent and phage resistance	24
	snp	4,894,057	1	"C" to "T" (Gln8 to stop codon, nosense mutation)	*ssg*	LPS capped with O antigen biosynthesis	less motility	26
	snp	5,605,723	1	"C" to "T" (Thr139 to Met)	*ampD*	beta-lactamase expression regulator	/	
	Deletion	6,069,210	18	"CGCCGCGCGCCGACCTG C" deletion in frame (deletion Pro193_Leu198)	*ampDh3*	beta-lactamase expression regulator	/	
	Deletion	6,228,926	1	"A" deletion in frame (frameshift mutation from Thr161)	*oprD*	outer-membrane porin that can facilitate the diffusion of specific substrates, such as basic amino acids and carbapenem antibiotics	oprD deficiency confers resistance to carbapenems, especially imipenem.	22

Table 2. Genomic changes in *P.aeruginosa* isolates compared to NPA09B.

However, the presence of the *ampR* mutation in NPA10B results in a decreased β-lactam resistance[3] reflecting the complex regulatory interactions between these mutations. NPA11B and NPA13B share a T83I mutation in *gyrA*, linked to quinolone resistance[26]and both have a frameshift mutation in the *nalC* gene, which induces overexpression of the MexAB-OprM efflux pump, contributing to increased β-lactam resistance[27]. These strains also harbor a mutation in the *oprD* gene, which leads to carbapenem resistance[28]. The combination of these mutations results in an increased MIC for levofloxacin, a slight decrease in MIC for β-lactams, and no change in MIC for carbapenems. Finally, NPA12B carries a frameshift mutation in the *migA* gene, associated with colistin resistance[29]. Overall, the MIC changes observed in these isolates are a result of the interplay between mutations in multiple genes regulating antibiotic resistance, including those affecting β-lactams, quinolones, carbapenems, and colistin.

Whole-genome sequencing analysis revealed the presence of genes associated with pyrimidine synthesis, LPS biosynthesis, and type IV pili assembly (Table 2). These functional genes primarily encode phage receptors and are also associated with bacterial virulence, motility and biofilm formation[30–32] To assess phenotypic changes in bacterial evolution during phage therapy, we further characterized the isolates for virulence traits, including motility, virulence, and biofilm formation capabilities to these isolates. We used *Galleria mellonella* larvae as an infection model, widely employed to study the virulence of both Gram-negative and Gram-positive bacteria[33,34]. We inoculated larvae of *Galleria mellonella* with five isolates of *P. aeruginosa*, including NPA09B, NPA10B, NPA12B, NPA11B, and NPA13B, as negative control PBS and positive control PAO1. After injection, we monitored the larvae's vitality to compare the virulence differences among the different isolates (Supplementary

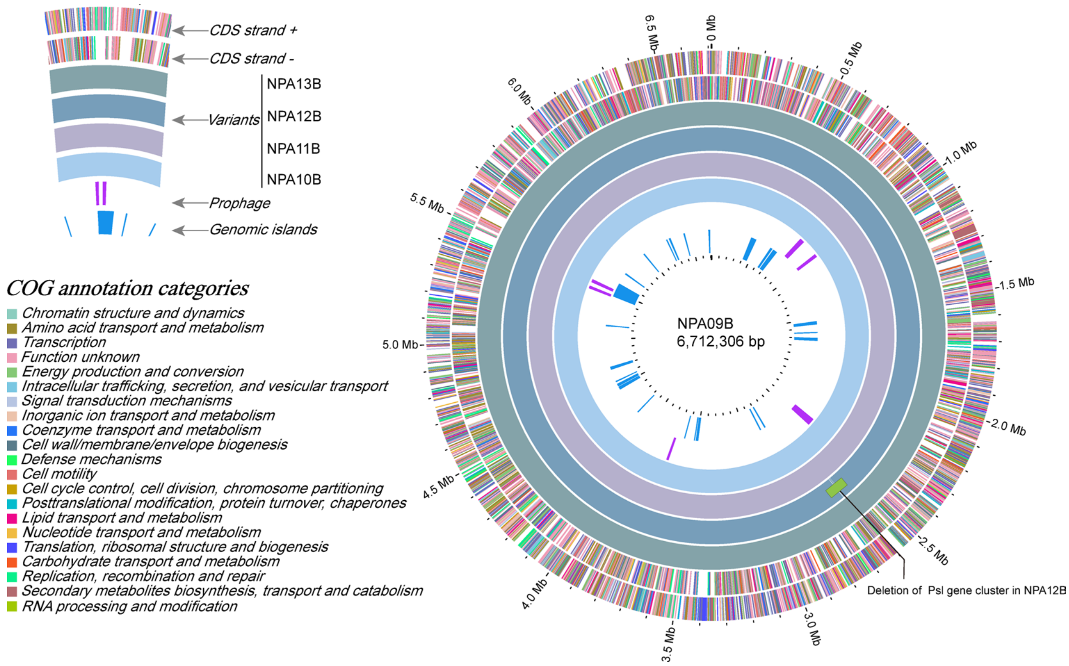

Fig. 3. Circular chromosomal view (CCV) of the bacterial genomes of five *Pseudomonas aeruginosa* isolates obtained just before (NPA09B) and during (NPA10B-NPA13B) phage therapy.

Fig. S2). The results showed that the virulence of NPA09B, NPA10B, and NPA12B was significantly higher than that of NPA11B and NPA13B (Fig. 4D). This could be related to the frameshift mutation in the *wapH* gene and a nonsense mutation in the *ssg* gene in NPA11B and NPA13B, which could potentially affect the synthesis of LPS O-antigen (Table 2). Previous studies have indicated that LPS O-antigen, as a critical component of the bacterial outer membrane, may influence bacterial virulence either by affecting phage adsorption or by participating in virulence-related mechanisms[31,32,35]. We also conducted twitching motility and phage adsorption experiments on the isolates. All strains, except NPA13B, exhibited proficient twitching motility (Fig. 4E), which may be associated with a mutation in the *pilQ* gene [36,37]. In the biofilm formation experiment, NPA12B exhibited the weakest biofilm formation ability among the strains, which is consistent with our whole-genome sequencing results. NPA12B had lost an 18 kbp Psl gene cluster associated with biofilm formation (Fig. 4F). NPA13B also showed decreased biofilm formation compared to the parent strain NPA09B (Fig. 4F), likely due to a mutation in the *dht* gene, as previous studies have shown that impaired pyrimidine metabolism can affect bacterial biofilm formation[38].

Discussion

Here, we report a case of personalized phage therapy for a patient with severe pneumonia caused by CRPA. Following CRPA infection, the patient exhibited life-threatening clinical symptoms, including coma and assisted ventilation. After receiving personalized phage therapy, the patient's condition stabilized. Although in the reported cases of phage therapy for *P. aeruginosa* infections, most were able to completely eradicate the bacteria[39,40]there are exceptions[18].

The development of phage-resistant strains during therapy poses a significant challenge in treating antibiotic-resistant infections. However, our study shows that phage-resistant strains do not necessarily prevent positive clinical outcomes. After the emergence of the phage-resistant *P. aeruginosa* strain NPA13B, we observed a reduction in the patient's bacterial load and an improvement in clinical symptoms, suggesting that phage therapy can still be effective despite resistance. Since we found that phage resistance in *P. aeruginosa* is associated with trade-offs in antibiotic resistance, virulence, motility, and biofilm formation (Fig. 4). And whole-genome sequencing confirmed these phenotypic changes, providing insight into how *P. aeruginosa* adapts to phage pressure (Table 2). This aligns with the "phage steering" hypothesis, which proposes that phage treatment can increase the susceptibility of bacteria to other antimicrobials or reduce their virulence, thus improving clinical outcomes[41]. Numerous studies have reported that bacteria often exhibit significant attenuation of virulence phenotypes during the acquisition of phage resistance. This phenomenon suggests that although bacteria may develop resistance to phages during phage therapy, such resistance development may come at the cost of reduced pathogenicity[42–44].

Genome analysis of the phage-resistant strains NPA11B and NPA13B revealed mutations potentially contributing to phage resistance. In NPA11B, a frameshift mutation in the *wapH* gene and a nonsense mutation in the *ssg* gene, both involved in the synthesis of LPS O antigen, likely impair phage attachment by altering the bacterial outer membrane, contributing to phage resistance[31,32]. In NPA13B, a frameshift mutation in *pilQ*, which is essential for Type IV pili formation, and an in-frame deletion in the *dht* gene, involved in pyrimidine

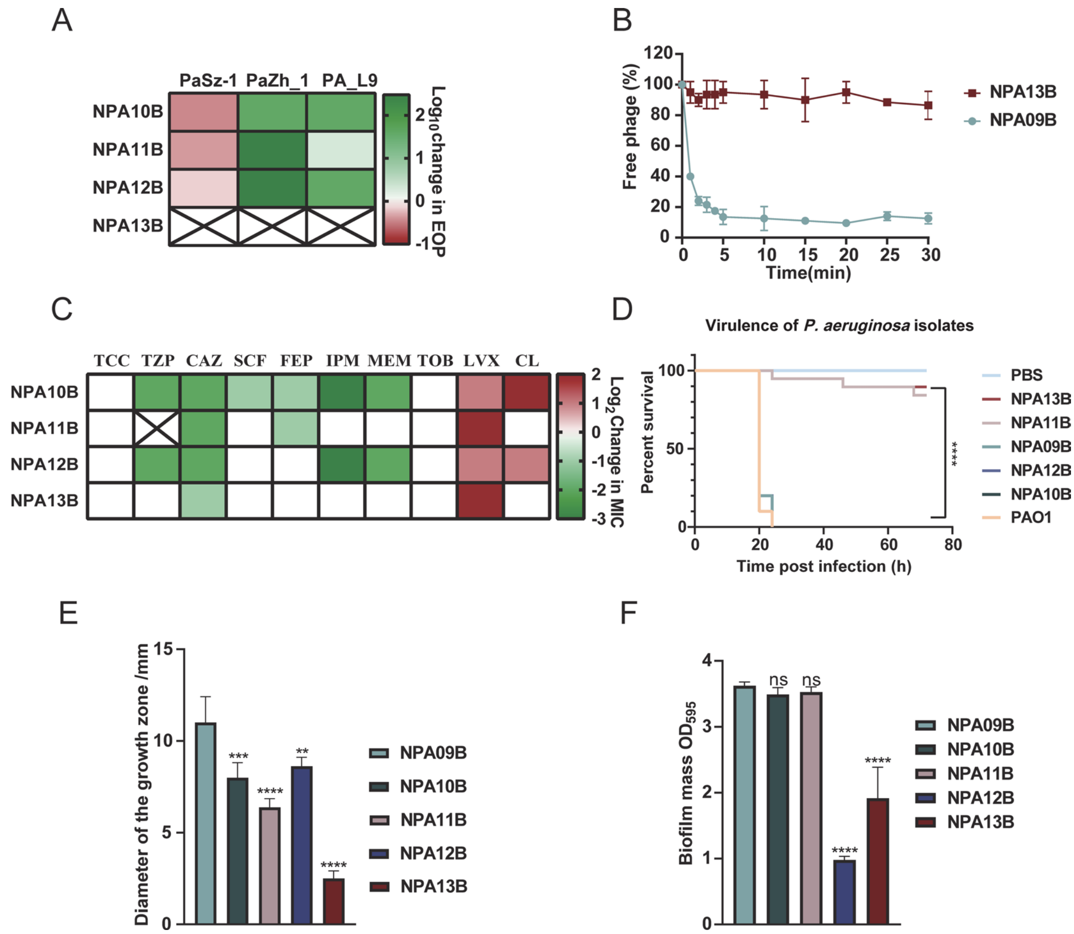

Fig. 4. Characterization of clinical strains of *P. aeruginosa*. (**A**) The determination of EOP (Efficiency of Plating) of phages on patient isolates. Red indicates a decrease in the EOP of the phage on that strain, and "×" means that the phage was unable to infect the strain. (**B**) Adsorption assay of PaSz-1_45_92k to NPA09B and NPA13B. (**C**) Change in antibiotic susceptibility of patient isolates, with green indicating regained sensitivity to antibiotics after phage therapy, "×" represents not determined. TCC, Ticarcillin/Clavulanic acid; TZP, Piperacillin/Tazobactam; CAZ, Ceftazidime; SCF, Cefoperazone/Sulbactam; FEP, Cefepime; IPM, Imipenem; MEM, Meropenem; TOB, Tobramycin; LEV, Levofloxacin; CL, Colistin. (**D**) Patient isolates were tested for their virulence in a *Galleria mellonella* infection model. The Kaplan–Meier plots represent the survival of 20 larvae that were used for each strain. The control group was injected with phosphate-buffered saline. Larvae survival was followed up to 72 h. The significance of larvae survival was assessed using a log-rank test, by comparison to the positive control strain NPA09B, "****" represents $p < 0.0001$. The survival curves of NPA09B and NPA12B are identical, as are the survival curves of NPA10B and PAO1, hence the graph only displays the survival curves for NPA09B and PAO1. (**E**) Tests of twitching motility in patient isolates, "**" represents $P < 0.005$; "***" represents $P < 0.0005$; "****" represents $P < 0.0001$. (**F**) Comparison of the biofilm-forming ability of patient isolates.

metabolism[45]were identified. The *pilQ* mutation likely disrupts phage binding[30]while the *dht* mutation may affect bacterial motility and biofilm formation, which can influence phage interactions[38]. While NPA13B showed complete resistance to the tested phages, these mutations in *ssg*, *pilQ*, and *dht* are likely responsible for the observed resistance by altering both surface structures and metabolic processes. However, further studies are required to fully elucidate the role of pyrimidine metabolism in phage resistance.

To address the challenge of bacterial resistance to phages, current research has proposed multidimensional synergistic intervention strategies to enhance therapeutic efficacy. Firstly, phage cocktail therapy, which combines multiple phages targeting different host receptors or exhibiting distinct lysis mechanisms, can effectively disperse bacterial selection pressure and significantly delay the emergence of resistant mutants[46]. Secondly, based on the complementary mechanisms of phages and antibiotics against bacterial survival pathways - for instance, phages disrupt bacterial structural integrity by lysing the cell wall, while antibiotics inhibit key metabolic processes such as protein synthesis - their combined use not only produces synergistic antibacterial effects, but preclinical studies have also demonstrated that this strategy can partially restore antibiotic sensitivity against resistant strains[15]. Furthermore, genetic engineering of phages has emerged as a breakthrough approach in recent years. By using CRISPR-Cas systems to precisely edit phage genomes, such as targeted knockout of host restrictionmodification system-related genes, insertion of lysin genes, or optimization of tail fiber protein-coding genes, this technology can simultaneously enhance lysing efficiency to improve bactericidal effects, expand host range to cover a broader spectrum of resistant strains, and even integrate lysin genes to strengthen targeted destruction of bacterial cell walls. Collectively, these strategies - ranging from multi-target interventions and pathway complementarity to genetic enhancement - form a comprehensive solution for combating phage resistance[47].

The exclusion of prophages from therapeutic preparations is a critical safety requirement for phage therapy, given their potential to harbor and transfer antibiotic resistance and virulence genes. Although a prophage-free host was unavailable for this emergency application, the final phage product was rigorously validated by WGS to confirm the complete absence of prophage contamination. Moving forward, a comprehensive safety strategy for clinical studies will involve the exclusive use of genomically verified, strictly lytic phages, propagation in engineered prophage-free hosts, and advanced purification techniques such as chromatography to ensure the absolute purity of the final product.

In the context of our patient's lung infection due to CRPA, we selected nebulized inhalation as the empirical route for phage administration. This choice was informed by the infection site's anatomical location and the need for direct and concentrated delivery of therapeutic agents to the affected area. Our approach aligns with the growing body of evidence supporting the use of aerosolized phage therapy for pulmonary infections, which suggests that local administration can enhance the therapeutic effect by ensuring a high concentration of active phages at the site of infection[48,49]. To substantiate the efficacy of our treatment approach, we monitored the titers of active phages in sputum samples and found that the concentration of phages reaching the site of infection was commensurate with the administered concentration. This finding is crucial as it validates the effectiveness of nebulized inhalation in delivering therapeutic phages directly to the infection site, thereby enhancing the potential for bacterial lysis and symptom alleviation. Multiple preclinical and clinical studies have demonstrated that aerosolized inhalation delivery can directly transport phages to pulmonary infection sites, significantly increasing local drug concentrations. Compared to intravenous injection, this administration method not only reduces systemic adverse reactions but also prolongs phage retention time on alveolar surfaces, thereby maintaining more sustained antibacterial activity[50].

Sepsis is defined as life-threatening organ dysfunction caused by a dysregulated response of the host to infection. During the patient's stay in the ICU, there was an elevation in blood urea nitrogen and creatinine levels, with almost zero urine output for two weeks, indicating symptoms of renal failure. According to the Sequential Organ Failure Assessment (SOFA) score, the patient's score was 10 [53]. Based on the severity of the condition assessed by the Acute Physiology and Chronic Health Evaluation II (APACHE II) scoring system[52]the patient's score was 26, with an estimated mortality rate of 56.93%. In the extremely critical condition of the patient, successful treatment with phages effectively controlled the further progression of the infection. Following cessation of colistin use, blood urea nitrogen and creatinine gradually decreased to normal levels, and urine output also returned to normal levels (Fig. 1H and K). These results demonstrate the safety and effectiveness of phage therapy.

Conclusion

Overall, the patient was infected with multidrug-resistant *P. aeruginosa*, resulting in life-threatening symptoms. Following personalized phage therapy, the patient's infection significantly improved, with normalization of inflammatory markers and improvement in lung lesions despite isolation of phage-resistant strains. While complete eradication of *P. aeruginosa* may not always be achievable, the reduced virulence of phage-resistant strains contributed to a marked improvement in the patient's clinical condition. This research emphasizes the importance of understanding the complex interactions between phages and bacteria to optimize phage therapy strategies.

Methods

Ethical approval and consent

This compassionate phage therapy was reviewed and authorized by the Southern University of Science and Technology Hospital Medical Ethics Committee, and the informed consent was obtained from the patient's direct relative.

Bacterial strains

The clinical isolates used in this study were obtained from routine microbiological cultures of clinical samples: sputum and bronchoalveolar lavage. All clinical isolates were stored in 20% glycerol at -80 °C. Antibiotic susceptibility of *P. aeruginosa* isolates was determined using the VITEK 2 system (bioMérieux). The classification of minimum inhibitory concentrations was based on clinical and laboratory association guidelines[53].

Phage isolation and purification

Phage isolation was performed according to the method reported by Chen et al.[54] with minor modifications. For phage enrichment, 2 mL of water sample, 1 mL of 3×Luria-Bertani (LB) broth and 200 µL of bacteria were mixed and incubated at 37 °C overnight. PaSz-1_45_92k, PaZh_1, and PAL9 Phages PaSz-1_45_92k and PaZh_1 were enriched and isolated using clinical isolate NPA09B as the host strain, whereas phage PAL9 was enriched and cultured with another clinical isolate, NPA10B, as the host. The waste water samples collected in our laboratory exhibit diverse origins, including medical effluents from healthcare facilities in Shenzhen, river water samples collected across multiple provinces in China, and aquaculture/agricultural wastewater discharged from farming bases. We performed phage isolation separately on these different waste water sources to ensure a broad host spectrum of the isolated phages and the generalizability of subsequent research.

The resulting cultures were centrifuged at 8000 g for 8 min. Next, 500 μL of phage enrichment solution, 3 mL LB and 200 μL of bacteria were mixed and incubated at 37 °C for the second round of enrichment. Phage enrichment solution was titrated on 0.7% LB agar containing host bacteria. Isolated phages were purified by triple streaking on 1.5% LB agar plates containing host bacteria. The purified phage was stored at 4 °C until required.

Phage preparation
Clinical strains were grown in LB broth, 30 mL of LB broth were inoculated with 300 μL of the overnight culture and incubated (220 rpm, 37℃) until early mid-exponential phase. In total, 100 μL of phage were added and the culture was incubated overnight at 37℃. The culture was transferred into a 50 ml centrifuge tube and centrifuged at 8 000 g for 8 min at 4℃. The supernatant was filter sterilized (0.22 μM) to remove the host bacteria. The phage preparation was purified in a cesium chloride density gradient as previously described[55] and then dialyzed using a Spectra/Por RC membrane (MWCO 20 K Da, Spectrum, New Jersey, USA) in phosphate-buffered saline to remove cesium chloride. Phages were then sterilized through 0.22 μM filters and titered on a lawn of clinical strains. Endotoxins were determined by the End-point Chromogenic Endotoxin Test Kit. Phage preparations were subsequently stored at 4 °C until required. Endotoxin levels of PaSz-1_45_92k, PaZh_1 and PAL9 were 1EU/mL, 2.31EU/mL and 9.98EU/mL.

Phage clinical therapy
Prior to clinical application, the phage preparation underwent quality testing at the Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences. Endotoxin levels were measured using the Limulus Amebocyte Lysate (LAL) assay and found to be below the FDA-specified limit of 5 EU/kg/h. Phage cocktail_1, composed of PaSz-1_45_92k and PaZh_1, was suspended in 1 ml of SM buffer (without tris), with each phage at a titer of 3×10^9 PFU/ml. After mixing, the titer of phage cocktail_1 was 6×10^9 PFU/ml and it was administered twice daily via an air jet nebulizer (Table 1). Baseline sputum samples were collected 1 h before the first phage therapy dose, followed by samples taken 2 h after each phage administration.

Determination of in vitro inhibition curves for phage/phage cocktail
For the clinical bacterial strains selected for phage screening, to further determine the bacteriolytic effect of the phages on drug-resistant bacteria, we separately determined the bacteriolytic curves of single phages and phage cocktails on the host bacteria. The host bacteria were cultured to an OD_{600} of approximately 0.5 (approximately 1×10^8 CFU/mL), then 100 μL of host bacteria were added to a 96-well plate, followed by the addition of 100 μL of 1×10^7 PFU of phages and phage cocktails. The plate was incubated at 37 °C with shaking, and OD_{600} absorbance values were measured every 1 h for a total of 12 h. Each experiment was performed in triplicate, and the results were analyzed using GraphPad Prism 8.

Phage DNA extraction
Precipitate the phage particles using 10% Polyethylene glycol 8000 at 4 °C overnight. Centrifuge at 10 000 g for 15 min and collect the pellet by removing the supernatant. Then suspend the pellet in SM buffer (100 mM NaCl, 8 mM MgSO4, 50 mM Tris-HCl), add 0.2 times the volume of chloroform, vortex mix, and let it stand for 10 min to allow phase separation. Centrifuge at 8 000 g for 10 min and collect the supernatant. Add 3 μL of DNase I and RNase A to treat the concentrated phage particles to remove bacterial nucleic acids. Extract the phage genomic DNA using the Lambda phage genomic DNA extraction kit.

Genome sequencing and analysis
The genomes of the five *P. aeruginosa* strains were sequenced as previously described[56]. Genomic DNA extraction was performed using a proprietary method developed by Guangdong Magigene Biotechnology Co.,Ltd. (Guangzhou, China). The integrity and purity of the DNA were assessed using a 1% agarose gel. Additionally, the DNA concentration and purity were measured using the Qubit 4.0 (Thermo Fisher Scientific, Waltham, USA) and NanoDrop One (Thermo Fisher Scientific, Waltham, USA), respectively. After the genomic DNA was qualified, according to the manufacturer's instructions, G-tubes were used to fragment the qualified genomic DNA and perform end-repair to prepare the SMRTbell DNA template library (selecting fragments above 10 kb using the BluePippin system)[57]. The library quality was evaluated using Qubit 4.0 (Thermo Fisher Scientific, Waltham, USA), and the average fragment size was evaluated using the Agilent 4200 (Agilent, Santa Clara, CA). SMRT sequencing was carried out on the Pacific Biosciences Sequel II sequencing instrument (PacBio, Menlo Park, USA) following the standard protocol[58]. Low-quality reads were filtered out, and the effective data after sample quality control were used for genome assembly. Pure third-generation data were assembled using Canu (v2.2) (https://canu.readthedocs.io/en/latest/quick-start.html), and the assembly results were optimized using the Arrow software, correcting any erroneous assembly regions.

Single nucleotide polymorphisms (SNPs) of the NPA09B, NPA10B, NPA11B, NPA12B, and NPA13B strains annotated by prokka v1.14.6[59] were identified using snippy v4.6.0. Large structural variations were examined using Geneious Prime. Visualization of the whole-genome map was performed with Proksee (https://proksee.ca/). Sequence typing was carried out with the mlst software v2.11. Further functional annotation, antibiotic resistance genes, prophage elements, and genomic islands were annotated using respectively eggNOG-mapper39 v2, CARD (Comprehensive Antibiotic Resistance Database), phigaro 2.3.0 (accessed in April 2024)[60] and IslandViewer4[61].

Counting of active phages in sputum samples
Take 500 μL of patient sputum and add 500 μL of Oxoid Sputasol digestion solution, prepared according to the instructions. Digest the patient's sputum at 37 °C until it becomes diluted. Centrifuge the mixture at 4 °C and 8 000 g for 5 min, then collect the supernatant. Dilute the supernatant tenfold and perform plaque-forming unit (PFU) counting, with each sample tested in triplicate.

Experiments on the virulence of ***Galleria mellonella***
The batch of *Galleria mellonella* larvae was purchased from Tianjin. The larvae, weighing 300 mg each, were stored at 15 °C in the dark before infection. Overnight bacterial culture was transferred at a ratio of 1:10 to fresh LB medium and incubated at 37 °C for approximately 2 h. OD_{600} was measured and colony-forming unit CFU was calculated. Bacterial suspensions were centrifuged at 5000 g for 5 minutes at 25 °C, washed with PBS, and the bacterial suspension concentration was adjusted to 10^5 CFU/ml. Using a 25 μL micropipette, 10 μL of the bacterial suspension was injected into the last left forelimb[62,63]. Wax moth larvae survival was monitored for 72 h, with larvae being considered dead if they darkened or showed no movement upon touch.

Adsorption experiment
The adsorption experiment was conducted following the protocol described by Kim et al., with the following modifications[64]. Exponentially growing host bacteria were infected with a phage suspension at a multiplicity of infection (MOI) of 0.01 and incubated at 37 °C. Samples of 200 μL were taken at 0, 1, 2, 3, 4, 5, 10, 15, 20, 25, and 30 min post-infection and immediately centrifuged at 12,000 g for 1 min. The supernatant was titrated using a double-layer agar method to determine the unadsorbed free phages.

Twitching motility assays
Twitching motility assays were performed as described by Sara L. N. Kilmury[65]. Briefly, strains of interest were stab inoculated to the bottom of an LB 1% agar plate with a P10 pipette tip, and plates were incubated upside down at 37 °C for 20 h. Following incubation, agar was carefully removed, and the plastic petri dish was stained with 1% crystal violet for 10 min. Excess dye was washed away with water and the diameter of the pumping zone. The one-way analysis of variance (ANOVA) statistical test was used to determine significant differences in twitching compared to WT.

The formation of biofilm experiment
Add 198 μL of culture medium and 2 μL of logarithmic phase bacterial suspension to each well of a 96-well plate, and incubate at 37 °C for 48 h. Remove the upper liquid, eliminate planktonic bacteria with 1×PBS through three washes. Then, add 200 μL of 0.1% crystal violet solution for staining for 5 min before aspirating the staining solution, followed by three washes with PBS. Air-dry the plate in a 37 °C incubator for 30 min. Add 200 μL of anhydrous ethanol to each well for dissolution and measure the OD_{595}.

Semi-quantitative culture scoring for patient's sputum
Cultures Samples were processed according to standard culture procedures, and the results were read after 48 h. The semi-quantitative scoring was determined by the four-quadrant method and classified as follows: 0 = no growth; + = rare growth; ++ = light growth; +++ = moderate growth; ++++ = heavy growth. Quantitative culture was performed by serial dilution of the respiratory samples. The colony counts were calculated by the number of colonies visible on the agar plate in relation to the dilution and inoculation factors[66].

Data analysis
Comparisons were performed by one-way ANOVA with Bonferroni's multiple comparisons test and Student's t-test. All statistical analyses were performed using Prism 8 (GraphPad, San Diego, CA, USA), and differences with $P < 0.05$ were considered statistically significant.

Data availability
All sequencing data generated in this study has been deposited in the NCBI BioProject database accession PRJN A1116835.(https://www.ncbi.nlm.nih.gov/bioproject/?term=PRJNA1116835).

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Author contributions

Y.Y., X.T., and M.X. contributed equally to this work. X.T., Y.M., and Y.P. designed the study. Y.Y. and X.T. wrote the main text. M.X., Z.L., S.L., H.G., R.G., J.Z., X.L., C.Z., S.W., X.D., Y.Z., and N.Z. acquired and analyzed the data. All authors have reviewed the manuscript and approved the submitted version.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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